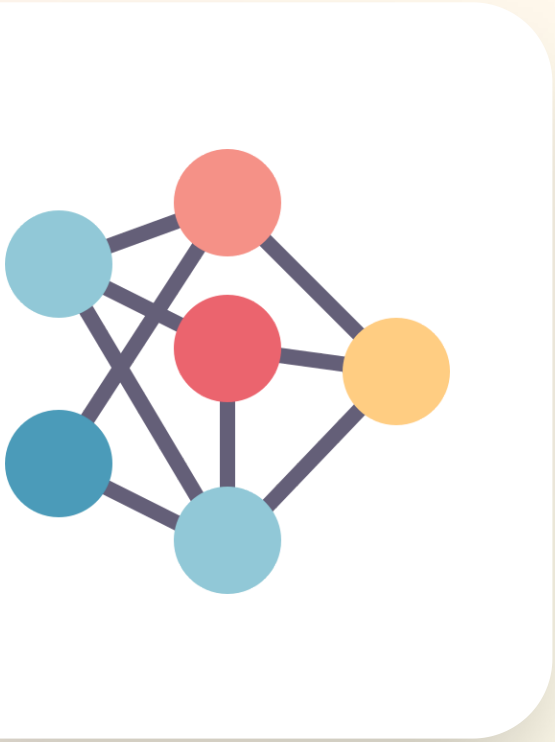
CHAPTER - 2

Candy Factory

The Four Factors

Addition • Subtraction • Multiplication • Division

"Practice these sums again and again to master them!"



Class 5 Mathematics

CANDY - 1

Addition Adventure

HOW TO SOLVE:

1. Write numbers vertically (Ones under Ones).
2. Start adding from the right (Ones column).
3. If sum is 10 or more, write the ones below and carry the tens to the TOP row.
4. Always add the carry digit first in the next column.

COMMON MISTAKES

- ✗ Aligning numbers from the left.
- ✗ Forgetting to add the carry digit at the top.
- ✗ Writing the whole sum in one column instead of carrying.

☆ Example: 9-Digit with Top Carries

- Align 568 and 985.
- 8+5=13. Write 3, Carry 1 to top.
- 6+8+1=15. Write 5, Carry 1 to top.
- 5+9+1=15. Write 15.

Final Answer: 1553



Example: Lakhs Alignment

- Align 479 and 498.
- 9+8=17. Write 7, Carry 1.
- 7+9+1=17. Write 7, Carry 1.
- 4+4+1=9.

Final Answer: 977



EXERCISE 1: STANDARD SUMS

1. 760051 + 822876
2. 832140 + 34510
3. 382115 + 662385
4. 147523 + 751070
5. 123263 + 678564
6. 977088 + 412570
7. 927484 + 858078
8. 158926 + 357963
9. 530088 + 888026
10. 841558 + 771466
11. 326955 + 326932
12. 540794 + 959829
13. 9495 + 383559
14. 704035 + 499166
15. 593438 + 649960
16. 888906 + 479104
17. 669870 + 287266
18. 533039 + 301086
19. 583622 + 190670
20. 747117 + 771703
21. 939145 + 896589
22. 798920 + 733007
23. 800188 + 589227
24. 521691 + 601767
25. 892640 + 220556

EXERCISE 2: ADDITION STORY PROBLEMS

1. A library had 45,600 books and added 12,400 more books. How many books are there in total?
2. Store A sold 5,00,000 chocolate bars and Store B sold 4,99,999 bars. What is the total number of bars sold?
3. The population of a town was 12,34,567 and it increased by 1,23,433 people. What is the new population?
4. A company earned ₹45,67,800 in January and ₹12,32,200 in February. What is the total earning?
5. NASA traveled 12 million km and ISRO traveled 15 million km. What is the total distance traveled?
6. A factory produced 1,25,000 toys in the morning and 2,75,000 in the evening. What is the total production?
7. A bank balance was ₹45,00,000 and ₹55,00,000 was deposited. What is the new balance?
8. A rocket used 1,20,000 liters of fuel and then used 80,000 liters more. What is the total fuel used?
9. Candidate A received 4,56,789 votes and Candidate B received 5,43,211 votes. What is the total number of votes?
10. A school bought 12,000 chairs and 8,000 desks. What is the total number of furniture items?

CANDY - 2

Subtraction Safari

HOW TO SOLVE:

1. Top number (Minuend) must be larger.
2. Start from right. If top digit < bottom digit, BORROW 1 from the next left.
3. Srike out the neighbor, decrease it by 1, and write the new value at the TOP.
4. Add 10 to your current digit. Subtract and repeat!

COMMON MISTAKES

- ✗ Forgetting to decrease the neighbor after borrowing.
- ✗ Subtracting smaller from bigger in reverse order.
- ✗ Confusion with multiple zeros (Borrowing chain).

☆ Example: Indian Style Borrowing

- Align 84,248 and 28,539.
- Ones: 8-9? Borrow from 4. 4 becomes 3. 8 becomes 18. 18-9=9.
- Tens: 3-3=0.
- Hundreds: 2-5? Borrow from 4. 4 becomes 3. 2 becomes 12. 12-5=7.
- Thousands: 3-8? Borrow from 8. 8 becomes 7. 3 becomes 13. 13-8=5.
- Ten-Thousand: 7-2=5.

	7	13	12	3	18
	~~8~~	~~4~~	~~2~~	~~4~~	~~8~~
-	2	8	5	3	9
	5	5	7	0	9

Final Answer: 55,709

Example: Borrow from Lakhs

- 50,00,000 - 12,50,000.
- Borrow across zeros. 5 becomes 4. Middle zeros become 9.
- Subtract normally.

Final Answer: 37,50,000

```
   4  9 10
   5  0 0,0 0, 0 0 0
 - 1  2 5,0 0, 0 0 0
 ---------------------
   3  7 5, 0 0, 0 0 0
```

EXERCISE 1: STANDARD SUMS (BORROWING)

1. 1729735 - 772799
2. 1886370 - 802955
3. 1803360 - 352905
4. 1773070 - 145327
5. 1014623 - 149689
6. 1388652 - 759876
7. 1781602 - 770241
8. 1892023 - 796404
9. 1993757 - 404549
10. 1728986 - 309252
11. 1068941 - 158413
12. 1656028 - 256666
13. 1029998 - 345204
14. 1924378 - 752861
15. 1717752 - 103273
16. 1509298 - 157863
17. 1095918 - 452365
18. 1782433 - 551212
19. 1234608 - 579895
20. 1396221 - 917968
21. 1071481 - 634641
22. 1360601 - 455892
23. 1904745 - 709940
24. 1400947 - 364519
25. 1628747 - 885986

EXERCISE 2: SUBTRACTION STORY PROBLEMS

1. A store had 10,00,000 candies and sold 7,50,600 candies. How many candies are left?
2. The total budget was ₹50,00,000 and ₹34,45,000 was spent. What is the remaining balance?
3. Town A had 45,000 people and 12,539 people moved away. What is the new population?
4. A journey is 5,600 km long and 3,450 km has been covered. How much distance is left?
5. A bank had ₹1,00,00,000 and ₹45,67,890 was withdrawn. What amount is remaining?
6. The maximum marks are 500 and a student scored 425. How many marks did the student miss?
7. A forest had 12,345 trees and 1,234 trees were cut down. How many trees remain?
8. A water tanker had 12,000 liters of water and 4,567 liters were used. How much water is left?
9. A company made a profit of 85 lakhs and had expenses of 60.5 lakhs. What is the net profit?
10. A student needs 1,200 points to qualify and has scored 845 points. How many more points are needed?

CANDY - 3

Multiplication Magic

HOW TO SOLVE:

1. Multiply by Ones first. No X needed.
2. Move to Tens. Add one 'X' placeholder at the end.
3. Move to Hundreds. Add two 'XX' placeholders.
4. Move to Thousands. Add three 'XXX' placeholders.
5. Add all the partial products together.

COMMON MISTAKES

- ✕ Forgetting placeholders (X).
- ✕ Wrong carry during multiplication.
- ✕ Skipping addition of rows.

☆ Example: 3-Digit Multiplier (X Logic)

- Multiply 2374 by 6 (Ones). Result 14244.
- Multiply 2374 by 2 (Tens). Add one 'X'. Result 4748X.
- Add 14244 + 47480 = 61724.

Final Answer: 61724

```
  2 3 7 4
x     2 6
---------
  1 4 2 4 4
+ 4 7 4 8 X
---------
  6 1 7 2 4
```

Example: Big multiplier (XX placeholders)

- 12345 x 125.
- Line 1: x 5 (Ones).
- Line 2: x 2 (Tens). Put 'X'.
- Line 3: x 1 (Hundreds). Put 'XX'.

Final Answer: 1,54,325

```
   12345
 x   125
-------
   61725
  24690X
 12345XX
-------
 15,43,125
```

EXERCISE 1: STANDARD SUMS (3-4 DIGIT MULTIPLIERS)

1. 1234 x 125
2. 4500 x 250
3. 8888 x 333
4. 5678 x 1001
5. 10101 x 99
6. 12345 x 111
7. 5000 x 250
8. 4321 x 123
9. 9000 x 450
10. 1111 x 1111
11. 25000 x 12
12. 7500 x 125
13. 1212 x 212
14. 4567 x 321
15. 1000 x 1234
16. 9876 x 205
17. 5555 x 444
18. 123 x 4567
19. 600 x 1200
20. 4444 x 222
21. 5000 x 111
22. 12345 x 12
23. 67890 x 10
24. 45000 x 30
25. 12000 x 12

EXERCISE 2: MULTIPLICATION STORY PROBLEMS

1. Each box contains 144 pens. If there are 12,000 boxes, how many pens are there in total?
2. A factory produces 4,500 items every hour. How many items are produced in 24 hours?
3. The price of 1 kg of apples is ₹120. What is the cost of 125 kg of apples?
4. One truck can carry 4,500 kg of goods. What is the total weight carried by 50 trucks?
5. Each notebook has 200 pages. How many pages are there in 5,000 notebooks?
6. The monthly rent is ₹12,500. What is the total rent for 2 years (24 months)?
7. The cost of one movie ticket is ₹250. If 450 people watch the movie, what is the total collection?
8. A student reads 45 pages every day. How many pages will the student read in 365 days?
9. One tree produces 125 apples. How many apples are produced by 8,000 trees?
10. A person saves ₹2,500 every month. How much will be saved in 120 months?

CANDY - 4

Division Discovery

HOW TO SOLVE:

1. Check how many times divisor fits in the first group.
2. Subtract to find remainder.
3. Drop down the next digit.
4. ZERO RULE: If you drop a digit and the number is STILL smaller than the divisor
5. you MUST put a 0 in the quotient before dropping another!

COMMON MISTAKES

- ✗ Forgetting to write 0 in quotient.
- ✗ Wrong subtraction.
- ✗ Skipping steps in long division.

Example: Zero Quotient Rule (5738 ÷ 19)

- 19 into 57 fits 3 times. Write 3.
- Subtract 57-57 = 0. Drop down 3.
- Can 19 fit into 3? NO. Put 0 in quotient.
- Drop down 8. Now it's 38. 19 x 2 = 38.
- Write 2. Final 302.

Final Answer: 302

```
      302
  19 | 5738
     - 57
      03
   (Drop 3, cant fit 19, keep 0)
      38
     - 38
       0
```

☆ Example: Division with Remainder (748 ÷ 6)

- 6 into 7 fits 1 time. Write 1.
- Subtract 7 - 6 = 1. Drop down 4 → 14.
- 6 into 14 fits 2 times. Write 2.
- Subtract 14 - 12 = 2. Drop down 8 → 28.
- 6 into 28 fits 4 times. Write 4.
- Subtract 28 - 24 = 4. This is remainder.

Final Answer: 124

```
      124
   6 | 748
     - 6
     ---
      14
    - 12
     ---
      28
    - 24
     ---
       4
```

EXERCISE 1: STANDARD SUMS (2-DIGIT DIVISORS)

1. 56,048 ÷ 12
2. 4,50,709 ÷ 15
3. 13,399 ÷ 13
4. 5,73,800 ÷ 19
5. 42,089 ÷ 40
6. 1,25,141 ÷ 25
7. 4,50,739 ÷ 50
8. 9,99,999 ÷ 11
9. 8,10,358 ÷ 90
10. 123456 ÷ 15
11. 568903 ÷ 20
12. 444,222 ÷ 12
13. 120,116 ÷ 30
14. 750,007 ÷ 15
15. 1,44,083 ÷ 17
16. 1,69,144 ÷ 13
17. 2,25,908 ÷ 24
18. 4,13,976 ÷ 18
19. 10,78,690 ÷ 24
20. 12,12,120 ÷ 26
21. 9,20,708 ÷ 27
22. 15,00,000 ÷ 25
23. 5,45,45,900 ÷ 50
24. 1,01,01,010 ÷ 11
25. 6,35,790 ÷ 15

EXERCISE 2: DIVISION STORY PROBLEMS

1. A factory made 9,000 candies and packed them equally into 30 boxes. How many candies in each box?
2. A school has 1,200 students. They are divided equally into 12 classes. How many students per class?
3. A shop earned ₹48,000 in 6 days. What is the earning per day?
4. A library has 15,000 books arranged on 50 shelves. How many books per shelf?
5. A truck carries 8,400 kg of goods in 7 trips. How much per trip?
6. A farmer harvested 6,300 fruits and packed them equally into 9 baskets. How many fruits per basket?
7. A company distributed ₹5,00,000 equally among 25 employees. How much does each get?
8. A train travels 3,600 km in 12 hours. What is the distance covered per hour?
9. A school bought 2,400 notebooks for 60 students. How many notebooks per student?
10. A tank contains 9,000 liters of water, used equally over 30 days. How much per day?

CANDY - 5

Decimals Discovery

The Values After Point

Tenths	Hundredths	Thousandths
1/10 or 0.1	*1/100 or 0.01*	*1/1000 or 0.001*

DECIMALS IDENTITY & QUIZ (20 QS)

1. What is 5/10 as decimal?
2. In 12.34, which digit is in Hundredths?
3. Write 'Six-Tenths' as a number.
4. Convert 0.007 into a Fraction.
5. Place of 9 in 0.129.
6. Write 1.5 in words.
7. Convert 45/100 into decimal.
8. In 0.876, name the position of 6.
9. Write 0.1 as a fraction.
10. Is 0.50 same as 0.5?
11. Write 'Twelve Thousandths' as number.
12. Convert 0.25 into fraction.
13. Place value of 4 in 4.5.
14. Place value of 4 in 0.4.
15. Compare: 0.1 [ ] 0.01
16. Write 9/1000 as decimal.
17. Name digit in Tenths place of 15.08.
18. Write 0.001 as fraction.
19. Decimal for 'Nine and five tenths'.
20. Convert 12/10 into decimal.

MIXED MARATHON!

25 STORY CHALLENGES FOR GENIUSES

FINAL 25 CHALLENGES

1. Town A: 45,67,120 ppl. Town B: 12,33,400. Combined?
2. 10,00,000 candies. 45,678 broken. Good count?
3. 1 book has 125 pages. Total in 5,000 books?
4. ₹1,25,000 shared by 25 winners. Each?
5. Spent ₹4,500 on 10 bags. Cost per bag?
6. Flight ticket ₹4,500. 20 people. Total?
7. 12 boxes of 50 chocolates each. Total?
8. Students: 1,200. Boys: 650. Girls?
9. Car travels 480km on 40L. km/L?
10. Bank had ₹1,00,00,000. Took ₹45,67,891. Left?
11. 15 rows, 25 chairs each. Total chairs?
12. 600 candies shared by 12 friends. Each?
13. Price of 1 gram gold ₹6,000. 100 grams?
14. A car travels 500km in 5 hours. Speed/hr?
15. Walked 1,200m then 800m. Total distance?
16. 12 pens per pack. 15 packs. Total pens?
17. 1,200 apples shared by 60 kids. Each?
18. Buy bike ₹75,000. Sell ₹62,000. Loss?
19. Movie bill ₹1,200 for 4 people. Each pays?
20. Price of 1kg apple ₹150. 10kg cost?
21. School has 1,200 students. Girls 40%. Count?
22. Spent ₹2,500 then ₹1,200 from ₹5,000. Left?
23. Library has 50k books. Gave 5k away. Left?
24. A forest has 1,200 trees. Each tree 50 fruits. Total?
25. Run 5km daily. Total in February (28 days)?